



Lethal impacts of insecticides and miticides on three agriculturally important aphid parasitoids

Kathy Overton^a, Samantha E. Ward^a, Ary A. Hoffmann^b, Paul A. Umina^{a,b,*}

^a Cesar Australia, Level 1, 95 Albert St, Brunswick, Victoria 3056, Australia

^b Bio21 Institute, School of BioSciences, The University of Melbourne, Victoria 3010, Australia

HIGHLIGHTS

- We tested the toxicity of 20 pesticides against 3 globally important aphid parasitoids.
- The aphelinid *Aphelinus abdominalis* showed greater tolerance than the two braconids species.
- Type I and II synthetic pyrethroid toxicity varied within and across families.
- We caution against making generalisations among parasitoid species and pesticide groups.

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ABSTRACT

Globally, agricultural industries are contending with escalating insecticide resistance evolution among a number of arthropod pests. The Australian grains industry is no exception, with the indiscriminate use of broad-spectrum insecticides contributing to the rise of resistant aphid populations. As such, there is a growing appreciation for the need to harness the full potential of biological control offered by aphid parasitoid wasps, yet few studies have quantified the non-target effects of pesticide use on these taxa using Australian populations and rates relevant to the grains industry. With this in mind, we followed IOBC protocols and tested the acute toxicity of 20 insecticides and miticides against three important aphid parasitoids: *Diaeretiella rapae* (M'Intosh), *Aphidius colemani* Viereck, and *Aphelinus abdominalis* (Dalman). We found toxicities across the three species were generally consistent for many of the active ingredients tested, however notable differences for paraffinic oil, diafenthiuron, indoxacarb, and thiodicarb were observed. Overall, the aphelinid, *A. abdominalis*, showed greater tolerance to several active ingredients, with lower IOBC toxicity rating categories compared with the two braconid species, *D. rapae* and *A. colemani*. Of the eight active ingredients tested that are currently registered to control aphids in Australian grain crops, only two (flonicamid and afidopyropen) had low toxicity across all three parasitoid species, with four active ingredients resulting in very high toxicity for one or more species. We additionally found that the toxicity of synthetic pyrethroids differed between chemical types, with type I pyrethroids (bifenthrin) found to be more toxic than type II (gamma-cyhalothrin and lambda-cyhalothrin) pyrethroids. Taken together, these results highlight the risks of making generalised toxicity inferences across active ingredients and species. We recommend further toxicity testing be undertaken in the field, in addition to quantifying the sub-lethal effects of insecticides and miticides on aphid parasitoids.

1. Introduction

Aphids (Hemiptera: Aphididae) are considered one of the most important insect pests in temperate climates (Minks & Harrewijn 1989; Guerrieri & Digilio 2008), causing significant losses in yield and/or quality in arable and horticultural crops (Dedryver et al. 2010). The four

processes aphids can impact host plants are: (1) damage incurred through feeding, (2) feeding-induced phytotoxicity, (3) transmission of viruses (with 46% of insect-borne viruses transmitted by aphids; Nault 1997), and (4) through production of filamentous saprophytic ascomycetes which can hinder photosynthetic activity (Dedryver et al. 2010). Controlling aphids within agricultural crops to prevent populations

* Corresponding author.

E-mail address: pumina@unimelb.edu.au (P.A. Umina).

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reaching economically damaging levels is therefore key in mitigating their potential impacts. Historically, pesticides (specifically broad-spectrum chemicals) have been utilised for aphid control (Dedryver et al. 2010). However, their indiscriminate use has contributed to the selection of resistant populations (e.g., green peach aphid *Myzus persicae* (Sulzer) and cotton aphid *Aphis gossypii* Glover; Dedryver et al. 2010; Pym et al. 2022). With 14 species of aphids known to have evolved insecticide resistance across one or more chemical Mode of Action (MoA) groups (Foster et al. 2017), the need to implement integrated pest management (IPM) programs to better manage aphid pests is crucial.

Biological control of pests via natural enemies (consisting of generalist predators and hymenopteran parasitoids) is a key component of IPM programs. Farmers can enhance existing populations of natural enemies through the conservation or inclusion of non-crop and resource-rich habitats (i.e., conservation biological control; Jonsson et al. 2008; Cullen et al. 2008; Gagic et al. 2018), or by augmenting predator and parasitoid populations through the release of commercially available species (i.e., augmentative biological control; Collier & van Steenwyk 2004; Crowder 2007; van Lenteren 2012), with demonstrable success in managing aphid pests (Skirvin et al., 2001; Seko et al., 2013; Jamont et al., 2014; Alarcón-Segura et al., 2022). However, the extensive use of pesticides in agroecosystems has lethal and sub-lethal impacts on non-target organisms (Devine & Furlong 2007), with insecticide and miticide use potentially triggering cascading impacts on invertebrate pests and natural enemy populations (Trumper & Holt, 1998; Stark et al. 2007). Selective pesticides, which are effective against target pests but reportedly have low to moderate effects on natural enemies, are increasingly being adopted, with their use over broad-spectrum products encouraged within an IPM context (Koss et al. 2005; Gentz et al. 2010; Varenhorst & O'Neal 2012; Machado et al. 2019).

Parasitoid wasps are among the most important biological control agents of pest aphids (Boivin et al. 2012). Globally, the acute and sub-lethal effects of insecticides and miticides on aphid parasitoids has been comprehensively studied (e.g., Desneux et al. 2005; Thies et al. 2005; Cloyd & Bethke 2011; Ohta & Takeda 2015; Garantonakis et al. 2016). However, there have been very few studies conducted on Australian populations of aphid parasitoids. Within Australia, there are dozens of aphid parasitoid species from two families, Braconidae and Aphelinidae (Carver and Stary, 1974; Waterhouse and Sands, 2001). The braconid parasitoids, *Aphidius* spp., *Lysiphlebus* spp., and *Diaeretiella rapae* (M'Intosh), and the aphelinid parasitoids, *Aphelinus* spp., are among the most abundant and effective parasitoids in Australian grains (Waterhouse and Sands, 2001; Heddle et al., 2020; Ward et al., 2021a,b).

The adoption of IPM programs to manage aphid pests in the Australian grains industry is hindered by a poor understanding of the impacts chemicals can have on biological control agents (Micic et al. 2008). While inferences can be made from studies conducted in other countries, there can be considerable differences in chemical responses between closely related species (Croft & Whalon 1982; Jansen 2000). This approach could therefore result in the under- or over-estimation of toxicity impacts. With this in mind, we undertook acute toxicity testing of 20 pesticides that are routinely used in the Australian grains industry on three economically important aphid parasitoids: *D. rapae*, *Aphidius colemani* Viereck, and *Aphelinus abdominalis* (Dalman). These species are generalist aphid parasitoids and commercially relevant, and are reared by several commercial insectaries globally. The aim of our study was to determine if inferences regarding insecticide and miticide toxicities can be generalised across aphid parasitoids, or if reported toxicities should be considered within a species-specific context.

2. Materials and methods

2.1. Parasitoids

Diaeretiella rapae, *A. colemani*, and *A. abdominalis* were provided as parasitised aphids (mummies) by the commercial supplier Biological

Services (Loxton, South Australia, Australia). Upon arrival, emerged adults were removed to ensure individuals of uniform age (<24 h old) were used in all experiments. The mummies were then placed in ventilated containers within a controlled temperature (CT) cabinet set at 20 ± 1.0 °C and a photoperiod of 16:8h (L:D), and left overnight, with a 10% honey solution supplied through a dental wick.

2.2. Insecticides and miticides

A detailed list of all the tested insecticide and miticide formulations, the rates tested relative to the Maximum Registered Field Rate (MRFR) and the standardised rate (g or mL a.i./ha) are shown in Table 1. Most insecticides and miticides were tested at 10% and 100% of the MRFR within Australian grain crops (APVMA, 2021). In the case of pirimicarb, dimethoate, and chlorpyrifos, we included two rates corresponding with the lowest and highest rates registered for use in Australian grain crops, as these three active ingredients are registered across a broad range of rates (Table 1). For nuclearpolyhedrosis virus (NPV), we tested toxicity at 100% and 1000% of the MRFR within Australian grain crops given the previously low reported toxicity towards aphid parasitoids (Wilkinson et al. 1975).

Due to the large number of treatments included in this study, toxicity testing was separated across two bioassays for each species. A negative control (deionised water) and a positive control (either chlorpyrifos or dimethoate) were included in all bioassays undertaken. Additionally, pirimicarb, tested at both the low (75 g a.i./ha) and high (500 g a.i./ha) rates, was included across all bioassays to ensure consistency of chemical responses. For *D. rapae*, thiodicarb was tested three times due to lower-than-expected mortality observed in the bioassays. Solutions of each chemical were made by diluting the necessary volume of product in deionised water.

2.3. Acute mortality bioassays

We assessed the acute toxicity of 17 insecticides and miticides commonly used in Australian grain crops via laboratory bioassays. The same methodology was used across all bioassays, which followed International Organisation for Biological Control (IOBC) protocols (Sterk et al., 1999), where parasitoids were exposed to dried spray residues. Each treatment was replicated 10 times, with each replicate consisting of three individuals within a petri dish (i.e., $n = 30$). In the negative control, we included an additional three replicates (i.e., 13 replicates in total; $n = 39$). A Potter Spray Tower (Burkard Manufacturing Co. Ltd., Rickmansworth, UK) was used to apply an even and consistent spray deposit in the base of a 60 mm diameter plastic petri dish (Corning, New York, US, 60 mm $\times$ 15 mm) for all bioassays, ensuring a deposit of 1 mg/cm². This spray deposit corresponded to a field application rate of 100 L/ha. After each petri dish was sprayed, it was immediately transferred into a fume hood, and residues left to dry for 30 – 60 mins.

Once dry, a 1 cm-long segment of dental wick was dipped in a 10% honey solution and placed on the lid of the petri dish to ensure that the food source would not be in contact with, and hence contaminated by, the chemical. Three adult parasitoids were then transferred onto the petri dish lid, and the petri dish base (with the dried residue) was inverted and placed on top of the lid to prevent escapees. Petri dishes were left unsealed to allow for ventilation within the exposure arena. For the sulfoxaflor 10% and 100% MRFR, spinetoram 10% MRFR, flonicamid 100% MRFR, and abamectin 100% MRFR treatments for *A. colemani*, and spinetoram 10% MRFR treatment for *A. abdominalis*, a single individual escaped from one replicate petri dish. To reflect this, the number of individuals from these treatments was $n = 29$. For the emamectin benzoate 10% MRFR treatment for *A. colemani* and the chlorantraniliprole 100% MRFR treatment for *A. abdominalis* treatments, an extra individual was placed into one of the replicate petri dishes and therefore the number of individuals within these treatments was $n = 31$.

Table 1

Mortality (%) observed for *Diaeretiella rapae*, *Aphidius colemani*, and *Aphelinus abdominalis* at 48 h after treatment (HAT) across 17 active ingredients. Most chemicals were tested at 10% and 100% of the Maximum Registered Field Rate (MRFR) in Australian grain crops, with exception to pirimicarb, dimethoate and chlorpyrifos, which were tested at the lowest and highest registered rates in Australian grains, and nuclearpolyhedrosis virus, which was tested at 100% and 1000% of the MRFR. Different letters indicate significantly different toxicities within species across all active ingredients ($p < 0.05$).

Active ingredient	Product formulation (g or mL a.i./kg or L)	Rate tested relative to the MRFR (%)	Standardised rate tested (g or mL a.i./ha)	Percent mortality (95% confidence intervals)		
				<i>Diaeretiella rapae</i>	<i>Aphidius colemani</i>	<i>Aphelinus abdominalis</i>
Water control - bioassay 1	–	10	–	0 (0–10.43) a	0 (0–10.43) a	0 (0–10.43) a
Water control - bioassay 2		100	–	3.03 (0.54–15.32) ab	0 (0–10.43) a	3.03 (0.54–15.32) ab
Nuclearpolyhedrosis virus ¹	7.5 × 10 ⁹ polyhedral inclusion bodies per mL	100	100	0 (0–11.35) a	0 (0–11.35) a	0 (0–11.35) a
		1000	1000	3.33 (1–17) ab	3.33 (1–16.67) ab	0 (0–11.35) a
<i>Bacillus thuringiensis</i> ²	280 ³	10	328.57	10 (3.46–24.62) ab	3.33 (0.59–16.67) ab	0 (0–11.35) a
		100	3285.7	0 (0–11.35) a	0 (0–11.35) a	0 (0–11.35) a
Chlorantraniliprole	350	10	2.45	0 (0–11.35) a	3.33 (0.59–16.67) ab	0 (0–11.35) a
		100	24.5	0 (0–11.35) a	3.33 (0.59–16.67) ab	0 (0–11.03) a
Flonicamid	500	10	10	0 (0–11.35) a	0 (0–11.353) a	0 (0–11.35) a
		100	100	3.33 (0.59–16.67) ab	3.45 (0.61–17.18) ab	3.45 (0.61–17.18) ab
Afidopyropen	100	10	0.5	3.33 (0.59–16.67) ab	0 (0–11.35) a	13.33 (5.31–29.68) ab
		100	5	6.67 (1.85–21.32) ab	3.33 (0.59–16.67) ab	3.33 (0.59–16.67) a
Abamectin	36	10	0.54	0 (0–11.35) a	30 (16.66–47.88) bc	0 (0–11.35) a
		100	5.4	23.33 (11.79–40.93) b	68.97 (50.77–82.72) de	20 (9.51–37.31) ab
Paraffinic oil	778	10	158.4	3.33 (0.59–16.67) a	16.67 (7.34–33.56) ab	6.67 (1.85–21.32) ab
		100	1584	23.33 (11.79–40.93) b	100 (88.65–100) e	3.33 (0.59–16.67) ab
Indoxacarb	150	10	6	0 (0–11.35) a	66.67 (48.78–80.77) cd	0 (0–11.35) a
		100	60	3.33 (0.59–16.67) ab	66.67 (48.78–80.77) cd	0 (0–11.35) a
Emamectin benzoate	17	10	0.51	6.67 (1.85–21.32) ab	25.81 (13.70–43.25) bc	0 (0–11.35) a
		100	5.1	53.33 (36.14–69.77) bc	86.67 (70.32–94.49) de	60 (42–75) cd
Diafenthiuron	500	10	30	3.33 (0.59–16.67) ab	3.33 (0.59–16.67) ab	20 (9.51–37.31) ab
		100	300	100 (88.65–100) d	60 (42–75) cd	46.67 (30.23–63.86) bc
Spinetoram	120	10	3.6	66.67 (48.78–80.77) bc	51.72 (34.43–68.61) cd	36.67 (21.87–54.49) bc
		100	36	100 (88.65–100) d	83.33 (66.44–92.66) de	65.52 (47.35–80.06) cd
Thiodicarb	375	10	28.125	6.67 (3.09–13.79) ⁴ ab	3.33 (0.59–16.67) ab	80 (62.69–90.49) e
		100	281.25	14.44 (8.64–23.16) ⁴ ab	93.33 (78.68–98.15) de	100 (88.65–100) e
Sulfoxaflor	500	10	5	93.33 (78.68–98.15) cd	100 (88.30–100) e	56.67 (39.20–72.62) cd
		100	50	100 (88.65–100) d	100 (88.30–100) e	86.67 (70.32–94.69) de
Pirimicarb	500	Low	75	100 (88.65–100) d	100 (88.65–100) e	96.67 (83.33–99.41) e
		High	500	100 (88.65–100) d	100 (88.65–100) e	96.67 (83.33–99.41) e
Methomyl	225	10	45	100 (88.65–100) d	100 (88.65–100) e	100 (88.65–100) e
		100	450	100 (88.65–100) d	100 (88.65–100) e	100 (88.65–100) e
Dimethoate	400	Low	30	100 (88.65–100) d	100 (88.65–100) e	100 (88.65–100) e
		High	320	100 (88.65–100) d	100 (88.65–100) e	100 (88.65–100) e
Chlorpyrifos	500	Low	35	100 (88.65–100) d	100 (88.65–100) e	100 (88.65–100) e
		High	750	100 (88.65–100) d	100 (88.65–100) e	100 (88.65–100) e

¹ The standardised rates reported for Nuclearpolyhedrosis virus are amount of product per Ha rather than amount of active ingredient.

² The standardised rates reported for *Bacillus thuringiensis* are amount of product per Ha rather than amount of active ingredient.

³ The concentration of active ingredient in the product formulation is 280 g a.i./kg, with 8640 IU of potency per mg of product.

⁴ Reported mortalities are based on testing conducted over three bioassays for *D. rapae*.

All petri dishes within a treatment were then placed in an 85 L sealed storage container (Inabox, Melbourne, Australia) to prevent desiccation and possible fumigant effects between chemical treatments. iButtons (Alphamach Inc., Quebec, Canada) were used to record temperature and relative humidity across multiple containers, which showed consistent temperatures ($21 \pm 1^\circ\text{C}$) and humidity ($97 \pm 2\%$) throughout all bioassays. Parasitoid mortality was assessed at 24, 48, and 72 h after treatment (HAT), with mortality assessed following IOBC protocols (Sterk et al., 1999; Brugger et al. 2010).

In addition to the 17 chemicals outlined in Table 1, we undertook further bioassays to specifically compare the responses of each parasitoid species to multiple synthetic pyrethroids. Here, we tested the toxicity of gamma-cyhalothrin, lambda-cyhalothrin, and bifenthrin following the methods described above. All three species were tested in one bioassay simultaneously to allow for comparison across active ingredients and species. For each chemical, we tested the MRFR in Australian grain crops, along with a negative control (deionised water) and a single positive control of dimethoate. Parasitoid mortality was assessed as outlined above.

2.4. Statistical analysis

All data was statistically analysed in the program R (Version 4.1.2). As parasitoid mortality in the control treatments was very low (between 0 and 3% throughout all bioassays), Abbott's correction was not applied to any of our data. For the primary bioassays which assessed the toxicity of the 17 insecticides, mortalities within each species derived across the two bioassays were pooled, with 95% confidence intervals (CIs) calculated using the functions 'biconf' and 'prop.test', to ensure reported CIs were consistent across the two functions. These 95% CIs were used to infer statistical significance between treatments; where 95% CIs failed to overlap, we assumed $p < 0.05$ (Greenland et al. 2016). We principally focussed on 48 HAT data in keeping consistent with IOBC protocols, although observed mortalities at 24 and 72 HAT can be seen in Tables S1 & S2. For the secondary bioassays assessing the toxicity of the synthetic pyrethroids, a binomial generalised linear model (function 'glm') was used to analyse the effect of species, active ingredient, exposure period (HAT), as well as an interaction of all three terms. The function 'anova' was used to perform an analysis of variance, where a binomial model with a logit link function was applied. Significance testing was performed across species and HAT, using the function 'emmeans' to conduct *post hoc* tests using 'sidak' for each active ingredient. Because thiodicarb was tested three times for *D. rapae*, we pooled the data across all bioassays to calculate a mean percentage mortality and calculated 95% CIs using this dataset.

In addition to the above, we assigned each chemical an IOBC toxicity rating (1 = <30%, 2 = 30–79%, 3 = 80–99%, 4 = >99% mortality; Sterk et al., 1999) based on mortalities observed at the MRFR at 48 and 72 HAT. This was performed for the three parasitoid species independently.

3. Results

3.1. Chemical responses among parasitoid species

At 48 HAT, the toxicities of NPV, *Bacillus thuringiensis*, chlorantraniliprole, flonicamid, and afidopyropen were low at both rates tested across all three species (<30% mortality; Table 1). While paraffinic oil at the MRFR had low toxicity for *D. rapae* (23.33% mortality) and *A. abdominalis* (3.33% mortality), it was highly toxic to *A. colemani* (100% mortality; Table 1). Similarly, at both 10% and 100% of the MRFR, abamectin had low toxicity towards *D. rapae* (0% and 23.34% mortality, respectively) and *A. abdominalis* (0% and 20% mortality, respectively), with moderate toxicities at both rates for *A. colemani* (30% and 68.97% mortality, respectively; Table 1). Indoxacarb elicited similar toxicity responses, with low mortalities observed for *D. rapae* and *A. abdominalis* (<4% mortality across both species at both rates),

although moderate toxicities were observed for *A. colemani* at 10% and 100% of the MRFR (66.67% mortality at both rates; Table 1).

For diafenthiuron, low toxicities were observed at 10% of the MRFR across all three species, but at the MRFR, moderate toxicities were observed for *A. abdominalis* (46.67% mortality) and *A. colemani* (60% mortality), and 100% mortality was observed for *D. rapae* (Table 1). Similarly, emamectin benzoate had low toxicity across the three species at 10% of the MRFR, although the MRFR was moderately toxic for *D. rapae* and *A. abdominalis* (53.33% and 60% mortality, respectively), and highly toxic for *A. colemani* (86.67% mortality; Table 1). We observed moderate toxicities across all three species for spinetoram at 10% of the MRFR, although at the MRFR, toxicity was moderate for *A. abdominalis* (65.52% mortality), highly toxic for *A. colemani* (83.33% mortality; Table 1), and very toxic for *D. rapae* (100% mortality; Table 1). Further, sulfoxaflor was particularly toxic to both *D. rapae* and *A. colemani* at 10% and 100% of the MRFR (93–100% mortality) but demonstrated lower toxicity to *A. abdominalis* (57–87% mortality; Table 1).

Similarly, thiodicarb demonstrated varying toxicities across species; at 10% of the MRFR, toxicity was low for *A. colemani* (3.33% mortality) but resulted in high toxicity at the MRFR (93.33% mortality; Table 1). Thiodicarb resulted in high toxicity at 10% of the MRFR for *A. abdominalis* (80% mortality) and was very toxic at the MRFR (100% mortality; Table 1). In contrast, thiodicarb demonstrated low toxicities at both 10% and 100% of the MRFR against *D. rapae* (6.67% and 14.44% mortality, respectively; Table 1).

Pirimicarb was very toxic at the 75 and 500 g a.i./ha rates towards *D. rapae* and *A. colemani* (100% mortality) and was also highly toxic to *A. abdominalis* (96.67% mortality at both rates; Table 1). We observed very high toxicities for methomyl, dimethoate, and chlorpyrifos across all three species at all rates tested (100% mortality in all cases; Table 1).

3.2. Differences in responses to synthetic pyrethroids

Mortality was significantly different across parasitoid species ($\chi^2 = 18.17$, $df = 2$, $p = 0.0001$) and across active ingredients ($\chi^2 = 490.79$, $df = 4$, $p < 0.0001$) at 48 HAT, although we did not detect a significant interaction between species and active ingredient ($\chi^2 = 14.11$, $df = 8$, $p = 0.079$). The type I synthetic pyrethroid, bifenthrin, was very toxic across all three species, with 100% mortality observed at 48 HAT (Fig. 1). In contrast, both type II synthetic pyrethroids, gamma-cyhalothrin and lambda-cyhalothrin, elicited a lower acute toxicity across the three species at 48 HAT. Gamma-cyhalothrin and lambda-cyhalothrin were moderately toxic towards *D. rapae* (66.67% and 63.33% mortality, respectively) but showed low toxicity towards *A. abdominalis* (0% and 13.33% mortality, respectively). For *A. colemani*, gamma-cyhalothrin had no toxic effect (0% mortality; Fig. 1), while lambda-cyhalothrin was moderately toxic (30% mortality; Fig. 1).

3.3. Changes in IOBC toxicity ratings with exposure period

Generally, IOBC toxicity rating categories within species remained the same between 48 and 72 HAT (Table 2). However, for some active ingredients and species, the additional exposure period of 24 h resulted in a shift in IOBC ratings. While most of these changes were an increase in a single category (i.e., 1 to 2, 2 to 3, or 3 to 4; Table 2), we found that the IOBC toxicity rating for indoxacarb increased from 2 (moderate toxicity) to 4 (very high toxicity) for *A. colemani* and increased from 1 (low toxicity) to 4 (very high toxicity) for *D. rapae* (Table 2).

4. Discussion

Our findings represent a comprehensive picture of chemical toxicity to key aphid parasitoids in Australia. In particular, the toxicity data generated for *D. rapae*, which is well recognised both within Australia (Waterhouse and Sands, 2001; Heddle et al. 2020; Ward et al. 2021b)

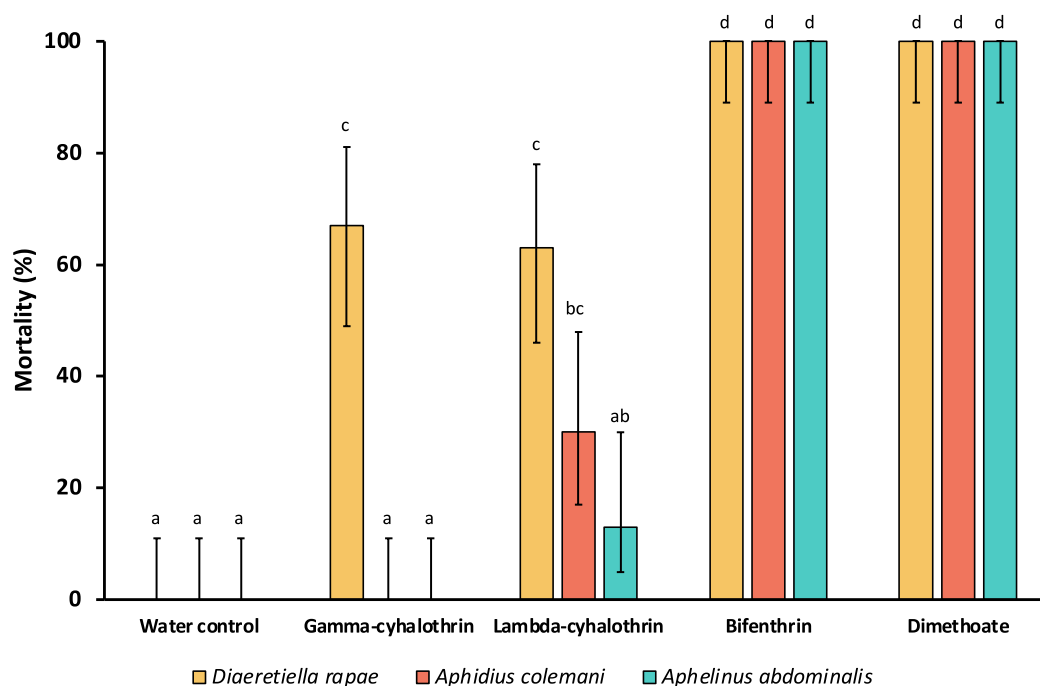


Fig. 1. Percent mortality observed for *Diaeretiella rapae*, *Aphidius colemani*, and *Aphelinus abdominalis* at 48 h after treatment (HAT) across three synthetic pyrethroids and a positive (dimethoate) and water (negative) control at the Maximum Registered Field Rate (MRFR) in Australian grain crops. Different letters indicate significantly different toxicities across species and active ingredients ($p < 0.05$).

Table 2

IOBC toxicity ratings (1 = <30%, 2 = 30–79%, 3 = 80–99%, 4 = >99% mortality) based on mortalities observed at the Maximum Registered Field Rate (MRFR) in Australian grain crops at 48 and 72 h after treatment (HAT) for *Diaeretiella rapae*, *Aphidius colemani*, and *Aphelinus abdominalis* across 20 active ingredients.

Active ingredient	<i>Diaeretiella rapae</i>		<i>Aphidius colemani</i>		<i>Aphelinus abdominalis</i>	
	48 HAT	72 HAT	48 HAT	72 HAT	48 HAT	72 HAT
Nuclearpolyhedrosis virus	1	1	1	1	1	1
<i>Bacillus thuringiensis</i>	1	1	1	1	1	1
Chlorantraniliprole	1	1	1	1	1	1
Fonicamid	1	1	1	1	1	1
Afidopyropen	1	1	1	1	1	1
Abamectin	1	2	2	3	1	2
Paraffinic oil	1	2	4	4	1	1
Indoxacarb	1	4	2	4	1	1
Gamma-cyhalothrin	2	2	1	1	1	1
Lambda-cyhalothrin	2	3	2	2	1	1
Emamectin benzoate	2	3	2	3	2	2
Diafenthiuron	4	4	2	3	2	2
Spinetoram	4	4	3	4	2	3
Thiodicarb	1	2	3	4	4	4
Sulfoxaflor	4	4	4	4	3	3
Pirimicarb	4	4	4	4	3	4
Methomyl	4	4	4	4	4	4
Dimethoate	4	4	4	4	4	4
Chlorpyrifos	4	4	4	4	4	4
Bifenthrin	4	4	4	4	4	4

and around the world (Pike et al. 1999; Desneux et al. 2006; Kant et al. 2012) as a significant generalist aphid parasitoid, is important, given the paucity of information in regard to effects of insecticides and miticides on this species (Overton et al. 2021). Toxicities observed for *D. rapae*, *A. colemani*, and *A. abdominalis* were consistent for many of the active ingredients tested, although notable species differences were detected. Additionally, synthetic pyrethroids led to different toxicities depending

on the type, with type I (i.e., bifenthrin) pyrethroids more toxic than type II (i.e., gamma-cyhalothrin and lambda-cyhalothrin). The variability in toxicity between type I and II synthetic pyrethroids has been previously demonstrated via a direct contact assay for the lepidopteran parasitoid *Colpoclypeus florus* (Walker), where permethrin (type I) and esfenvalerate (type II) sprayed at their recommended rates resulted in 100% and 72% mortality, respectively (Brunner et al. 2001). Our findings highlight the importance of incorporating both type I and II synthetic pyrethroids when testing chemical toxicity; something that is not often undertaken in published studies on beneficial arthropods (e.g., Garzón et al. 2015; Ohta & Takeda 2015; Koch et al. 2020). These results also demonstrate the risks of making generalised toxicity inferences across pyrethroid compounds.

Of the 20 active ingredients tested here, eight are currently registered for use against aphids in Australian grain crops: dimethoate, fonicamid, afidopyropen, gamma-cyhalothrin, lambda-cyhalothrin, paraffinic oil, pirimicarb, and sulfoxaflor (APVMA, 2021). These chemicals are also widely used to control aphid pests in many other countries around the world. Of these, fonicamid and afidopyropen were the only active ingredients that demonstrated low toxicity across all three aphid parasitoid species. Jansen et al. (2011) also found low toxicity when *A. colemani* was exposed to fonicamid via laboratory and field treated plants (5% and 14% mortality, respectively), although higher toxicity (44% mortality) was found when *A. colemani* was exposed to residues on a glass plate. While Koch et al. (2020) has previously reported a moderate toxicity (63% mortality) 7 days after exposure to afidopyropen for *Aphelinus certus* Yasnosh, the rate tested was almost double than what we tested (i.e., 9.78 g a.i./ha). Paraffinic oil had very low toxicity against *A. abdominalis*. However, paraffinic oil resulted in 100% mortality against *A. colemani*, and while we found low toxicity at 48 HAT for *D. rapae* (23% mortality), we observed moderate toxicity at 72 HAT (70% mortality). Although often marketed as an IPM compatible product, paraffinic oils have shown high toxicity against aphid parasitoids in other studies. For example, Satar et al. (2012) found 97%, 100%, and 96% mortality when paraffinic oils were sprayed at a rate of 2400 g a.i./ha on *Lysiphlebus confusus* Tremblay & Eady, *Lysiphlebus fabarum* Marshall, and *Lysiphlebus testaceipes* (Cresson),

respectively. However, this contrasts with findings by Urbaneja et al. (2008), who reported a low toxicity (12% mortality) when *A. colemani* adults were exposed to 1912.5 g a.i./ha of paraffinic oil in a residual contact assay.

Pirimicarb, sulfoxaflor, and dimethoate were highly toxic towards all three species of aphid parasitoids. The very high toxicity of pirimicarb at both rates tested (i.e., 75 and 500 g a.i./ha) was somewhat unexpected, given that pirimicarb is often marketed, and frequently discussed in the literature, as being selective (e.g., Theiling & Croft 1988; Bacci et al. 2009; Jansen et al. 2011). However, consistent with our results, Jansen (1996) found pirimicarb to be highly toxic to *Aphidius rhopalosiphii* DeStefani-Perez when exposed to residues sprayed at a rate of 125 g a.i./ha. Furthermore, the toxicity of pirimicarb has been observed in a field trial in wheat, where parasitic Hymenoptera significantly decreased after plots were sprayed with 140 g a.i./ha, reportedly due to a combination of toxicity and host removal (Powell et al. 1985). The very high toxicities we observed following sulfoxaflor and dimethoate exposure were also consistent with previous studies. For sulfoxaflor sprayed at similar rates to those used here, 100% mortality was reported across multiple species of *Aphidius* after direct and residual contact (Kim et al. 2018; IOBC Database, 2022). Similarly, the broad-spectrum effect of dimethoate has been observed across *Aphidius* and *Aphelinus* species, with 100% mortality consistently reported after exposure (e.g., Hassan et al., 1988; Hautier et al., 2006; Stara et al., 2011; Frewin et al., 2012). Although gamma-cyhalothrin and lambda-cyhalothrin were low to moderately toxic towards the three species tested here, it is worth noting these chemicals have been found to be very toxic for a range of other natural enemies of aphids, including lacewings and ladybird beetles (Booth et al. 2007; Pezzini & Koch 2015).

In terms of IOBC toxicity ratings, we found that ratings were similar across most active ingredients for both braconid species, *A. colemani* and *D. rapae*, although some differences were observed at the MRF for paraffinic oil, abamectin, diafenthiuron, indoxacarb, spinetoram, gamma-cyhalothrin, and lambda-cyhalothrin. These two species are widely distributed and contribute to high rates of field parasitism in Australia (Heddie et al. 2020; Ward et al. 2021b). Overall, the aphelinid, *A. abdominalis*, showed greater tolerance to many of the active ingredients tested, with lower IOBC toxicity rating categories compared with the two braconid species. The inconsistent toxicities observed between the two families could be explained because of behavioural (e.g., frequency of antennae cleaning and active searching behaviour; McGregor & Mackauer 1989) or physiological differences, perhaps reflecting a history of exposure of the reared population to insecticides and miticides (c.f. Desneux et al. 2004). Differing chemical susceptibilities in species within the same family have also been reported previously. Prabhaker et al. (2007) demonstrated variability in pesticide toxicity across four species of parasitoids from two Hymenopteran families, Aphelinidae and Mymaridae. Among the three Aphelinidae species tested, *Aphytis melinus* Debach was more susceptible to seven different insecticides compared with *Eretmocerus eremicus* Rose & Zolnerowich, and *Encarsia formosa* Gahan. Similarly, Tillman & Mulrooney (2000) reported differences in susceptibility among three braconid species, *Bracon mellitor* Say, *Cardiochiles nigriceps* (Say), and *Cotesia marginiventris* (Cresson). For example, lambda-cyhalothrin was found to be highly toxic to *B. mellitor* and *C. nigriceps*, but less toxic to *C. marginiventris*. In contrast, spinosad demonstrated low toxicity across all three braconids (Tillman & Mulrooney 2000). Combined with our study, these results demonstrate the risk of making generalisations about acute toxicity within and across natural enemy families and highlight the need for further research to understand the underlying behavioural and physiological differences that contribute to the variability in toxicity responses. This mirrors the findings of Banks et al. (2011), who also caution against using 'representative' (or 'surrogate') species when making generalisations about chemical toxicity within guilds.

We followed IOBC protocols when conducting toxicity testing (Sterk et al., 1999), however we additionally included mortality assessments of

parasitoids at 72 HAT. While IOBC toxicity rating categories remained consistent between 48 and 72 HAT for many active ingredients within each species, there were several instances where ratings increased with longer exposure. Although unsurprising, these results point to limitations of IOBC protocols and suggest that mortality assessments at later times may be useful. This was particularly evident for indoxacarb, which demonstrated a delayed toxicity response in both *D. rapae* and *A. colemani*. At 48 HAT, mortality was 3% for *D. rapae* (i.e., low toxicity) and 67% for *A. colemani* (i.e. moderate toxicity), which increased to 100% mortality (i.e. very high toxicity) at 72 HAT for both species. The delayed toxicity of indoxacarb has previously been noted for *A. colemani*, where Stara et al. (2011) observed an increase in mortality from 24% at 24 HAT to 95% at 48 HAT. Similarly, Bostanian & Akalach (2004) found a delayed effect across three rates of indoxacarb, with mortality consistently increasing between 24 HAT and 216 HAT.

Although we have addressed important knowledge gaps for *D. rapae*, *A. colemani*, and *A. abdominalis*, we recommend additional testing is conducted to validate our findings through semi-field and/or field trials (Stark et al. 1995; Stark et al. 2007). As discussed in Jansen (2001), the artificial environment created by the testing arenas used in laboratory studies, and the contact between treated surfaces and aphid parasitoids, is forced and unnatural. If our study was conducted under more representative conditions found in crops, toxicities may vary from those observed here, given that adult parasitoids are able to modify their behaviour via the detection of pesticide residues which can have repellency effects (Kühner et al. 1985; Jansen 1999; Jansen 2000). Furthermore, the toxicities observed in our study are likely to vary if conducted under different environmental conditions (i.e., temperature, humidity, UV exposure; Holmstrup et al. 2010; Abbes et al. 2015), which should be considered when conducting field validation studies. Further, given that our bioassays were conducted using three commercially reared and sourced populations with unknown pesticide exposure histories, our findings may not be broadly applicable to all field populations. Additionally, our study did not investigate potential sub-lethal effects of insecticide and miticide exposure. Despite low toxicities observed following exposure to some active ingredients, reproduction, fecundity, sex-ratios, as well as other intergenerational effects are likely to be impacted in some way, as has been recorded in other studies (e.g., Desneux et al. 2007; Mardani et al. 2016). We recommend that further testing using Australian populations of these three species is conducted to quantify potential sub-lethal effects, particularly for active ingredients where low or moderate toxicity has been observed. Finally, while we tested a total of 20 active ingredients, these were limited to insecticides and miticides, and hence did not incorporate any herbicides or fungicides. Although arguably less important than insecticides and miticides in this field of research, herbicides and fungicides can also lead to non-target effects on natural enemies (Jansen 1999).

CRedit authorship contribution statement

Funding acquisition: AAH, PAU; Conceptualization: KO, AAH, PAU; Methodology: KO, SEW, PAU; Formal analysis: KO, AAH; Writing – original draft: KO; Writing – review & editing: KO, SEW, AAH, PAU.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biocontrol.2022.105143>.

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